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Developing intelligent-TPACK (I-TPACK) framework from unpacking AI literacy and competency: implementation strategies and future research direction

Introduction

The rapid integration of Artificial Intelligence (AI) into education – marked by generative AI (e.g. ChatGPT, Midjourney), adaptive learning platforms, and data-driven analytics – has fundamentally reshaped teaching and learning paradigms. AI offers transformative potential for personalised instruction, automated assessments, and enhanced student engagement. However, its integration remains uneven due to a critical gap: the absence of robust frameworks guiding teachers in effectively harnessing these tools (Chiu, 2024; Li et al., 2024). Existing frameworks like the Technological Pedagogical Content Knowledge (TPACK) framework offer fundamental ideas about teacher technology integration but fall short in addressing AI's unique capabilities and challenges, such as autonomous content generation, ethical dilemmas, and dynamic human-AI collaboration (Chiu & Rospigliosi, 2025). This gap underscores the urgent need to bridge AI literacy – defined as a foundational understanding of AI concepts, ethical implications, and critical evaluation skills – and AI competency, which emphasises the confident, reflective, and ethical application of AI in pedagogical practice (Chiu, 2025a, 2025b). Without this synergy, teachers risk underutilising AI's potential or exacerbating issues like bias, misinformation, and the erosion of student critical thinking.

To advance scholarly dialogue on these critical issues and research trajectories, this editorial proposes the Intelligent-TPACK (I-TPACK) framework – an expansion of Mishra and Koehler's (2006) Technological Pedagogical Content Knowledge (TPACK) framework that integrates domain-specific AI dimensions. I-TPACK reconceptualizes teachers' roles through novel intersections while emphasising essential competencies in validation, prompt engineering, and human-AI collaboration.

Our proposed I-TPACK framework

Traditional TPACK – while robust for inert digital tools – fails to account for AI's agentic autonomy (e.g. generative content creation), dynamic adaptability (e.g. real-time scaffolding), and socio-ethical entanglements (e.g. algorithmic bias). Our proposed I-TPACK framework emerges as a critical response to the transformative impact of AI on educational paradigms. This framework bridges this gap by integrating AI literacy (technical and ethical understanding of AI systems) and AI competency (applied, ethical implementation) as foundational pillars, positioning teachers as augmented pedagogy designers and instructors who strategically orchestrate human-AI synergies. Grounded in the ideas of sociotechnical pedagogy (Schmidt et al., 2024), the human-AI collaboration model (Chiu & Rospigliosi, 2025), and the responsible digital citizen (Chiu, 2025b), the I-TPACK reconceptualises TPACK to address AI's unique epistemic, pedagogical, and ethical complexities. Sociotechnical pedagogy transforms TP, CK, and PK into AI-PT, AI-CK, and AI-PK, respectively; the human-AI collaboration model encourages human-AI collaborative knowledge (HAIC-K); and the responsible digital citizen advocates ethical knowledge (Ethic-K). Therefore, our proposed I-PACK framework has five knowledge domains; see Figure 1.

- **Sociotechnical pedagogical reconceptualisation:** AI tools are not passive instruments but active agents that reshape knowledge construction. I-TPACK thus redefines teacher agency: teachers transition from content deliverers to validators and contextualisers of AI-generated knowledge, ensuring epistemic integrity while leveraging AI's scalability. This transforms TP, PK, and CK into AI-TK, AI-PK, and AI-CK, respectively.
 - **AI-Technological Knowledge (AI-TK)** refers to teachers' mastery of AI systems' operational logic, capabilities, and limitations. This knowledge domain goes beyond basic digital literacy to include prompt

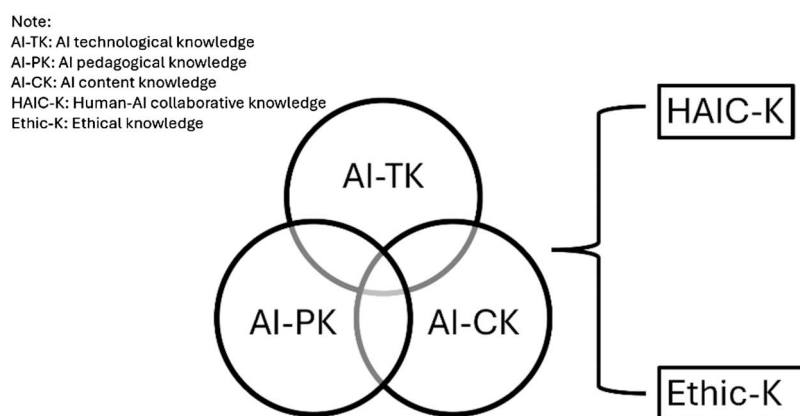


Figure 1. I-TPACK framework.

engineering (strategically crafting inputs to optimise AI outputs), algorithmic awareness (understanding data dependencies and model biases), and system auditing (evaluating reliability, security, and ethical compliance). For example, a teacher leverages AI-TK when designing ChatGPT queries to generate differentiated discussion prompts while proactively mitigating hallucinations through constrained parameters. AI-TK enables teachers to harness AI's potential while navigating its technical constraints.

- **AI-Content Knowledge (AI-CK)** refers to the ability to critically curate, validate, and contextualise AI-generated disciplinary content. It requires epistemic vigilance (triangulating AI outputs against authoritative sources), bias detection (identifying algorithmic inaccuracies in training data), and pedagogical contextualisation (aligning AI content with curriculum standards and real-world relevance). For example, a mathematics teacher exercises AI-CK by cross-referencing calculus solutions with geometric proofs, then leading discussions on computational limitations. This knowledge domain ensures AI serves as a validated resource rather than an unchallenged authority.
- **AI-Pedagogical Knowledge (AI-PK)** refers to the ability to strategically design AI-augmented learning experiences through metacognitive orchestration, which involves blending AI tools with human instruction; assessment redesign, which focuses on measuring skills that AI cannot replicate, such as creativity; and scaffolded collaboration, which emphasises structuring human-AI co-agency. For example, a language teacher incorporates ChatGPT-generated essay drafts into peer-learning lessons to demonstrate AI-PK. Students refine their arguments while documenting the influence of AI on their rhetorical voice. AI-PK transforms teachers into augmented pedagogy designers who ethically align technology with pedagogical objectives.
- **Human-AI Collaborative Knowledge (HAIC-K):** The human-AI collaboration model (Chiu & Rospigliosi, 2025) asserts that learning efficacy is maximised when teachers, students, and AI coordinate cognitive tasks. I-TPACK formalises this through human-AI collaborative knowledge that structures interactions where AI handles procedural tasks (e.g. grammar correction), freeing teachers for higher-order mentorship.
- **Ethical Knowledge (EK):** Unlike TPACK's peripheral treatment of ethics, I-TPACK centralises it as a core domain. This acknowledges AI's inherent moral and ethical dimensions – from data privacy to algorithmic justice – requiring explicit pedagogical attention.

Knowledge domain interdependencies

The five knowledge domains that are provided are interdependent and must be coordinated to facilitate the effective integration of AI into the teaching process for teachers. For example,

- **AI-TK enables AI-CK:** Understanding model limitations (AI-TK) is a prerequisite to validating AI-generated content (AI-CK). For example, A teacher understands that Photomath solves equations via symbolic computation (not conceptual reasoning) and may skip alternative methods (AI-TK); teachers should prioritise conceptual clarity (e.g. uses quadratic formula without factoring) over efficiency over

- AI-PK mediates HAIC-K: Pedagogical designs (AI-PK) determine how AI integrates into group dynamics (HAIC-K). For example, teachers design a peer-review workflow (AI-PK) in which AI can take on roles such as co-author, silent observer, and critical peer (HAIC-K).
- Ethic-K permeates all domains: ethical auditing (e.g. bias detection in AI-TK) informs content validation (AI-CK) and collaboration protocols. Ethic-K Trigger: Teachers should audit an AI application before using it for teaching.
- HAIC-K structures AI-PK: HAIC-K structures how AI participates in learning (e.g. as peer reviewer/data analyst), while AI-PK determines when and why these roles shift dynamically based on pedagogical goals. For example, AI acts as a simulation partner generating hypotheses (HAIC-K) while the teacher pauses AI interaction to facilitate small-group debate on algorithmic assumptions (AI-PK).
- Ethic-K governs AI-CK: Ethic-K provides normative frameworks (e.g. fairness protocols) that govern (AI-CK)'s validation processes. For example, before validating the pollution impact analysis generated by AI (AI-CK), the teacher applies Ethic-K protocols: understand what data was used for training. Without Ethic-K, AI-CK validation may overlook ethical and moral issues.
- AI-TK facilitates HAIC-K: AI-TK (e.g. prompt engineering) allows real-time role reconfiguration in HAIC-K workflows. For example, in AI-TK, teachers use prompts to shift their roles from writing assistant to devil's advocate, while students engage in dialectical discourse with the reconfigured AI in HAIC-K.
- Ethic-K monitors AI-PK: AI-PK's assessment redesign (e.g. evaluating creativity) is constrained by Ethic-K's equity principles, while ethical dilemmas encountered in practice refine Ethic-K frameworks. For example, no AI tools appropriately measure socioemotional skills (e.g. perseverance); teachers need to design peer reflections + AI analytics hybrid assessments (AI-PK).

Our proposed I-TPACK framework resolves four key limitations in traditional TPACK: (i) Tool agency: AI's autonomy necessitates AI-TK (vs. TK's focus on static tools); (ii) Epistemic challenge: AI's content generation demands AI-CK for critical validation; (ii) Ethical complexity: Algorithmic risks require embedded Ethic-K; (iv) Dynamic collaboration: AI's interactive role formalises HAIC-K.

Implementation strategies for the I-TPACK framework

To translate the Intelligent-TPACK (I-TPACK) framework into practice, a multi-tiered approach targeting teacher development, curriculum design, institutional infrastructure, and assessment reform is essential. We suggest three approaches.

- Teacher competency-based upskilling: Effective implementation begins with job-embedded professional development that targets I-TPACK's synergistic domains. Micro-credentials should scaffold AI-TK → AI-CK pathways – training teachers to audit tool limitations (e.g. Photomath's omission of geometric proofs) before validating content – and AI-PK → HAIC-K integration, where educators design lessons with dynamic AI role shifts. Crucially, Ethic-K infusion must precede tool deployment through bias-detection drills. Workshops should simulate classroom scenarios using AI platforms, enabling teachers to practice prompt engineering, validation protocols, and ethical troubleshooting.
- Curriculum integration: I-TPACK requires domain-specific pedagogical architectures that operationalise knowledge interdependencies. For example, in mathematics, teacher training on AI validation might sequence (1) problem-solving via Photomath (AI-TK), (2) geometric cross-verification of solutions (AI-CK), and (3) ethical reflection on algorithmic gaps (Ethic-K). For language arts, "Rhetorical Co-Authorship" units could involve (1) draughting with Gemini (HAIC-K), (2) peer-editing for human voice (AI-PK), and (3) transparency statements disclosing AI contributions (Ethic-K). Schools should curate repositories of vetted prompt libraries and bias-audited tools, categorised by subject and competency level.
- Collaborative inquiry networks: Establish school-based or teacher-led I-TPACK Labs where interdisciplinary teacher teams co-design, test, and refine AI-augmented pedagogies through cyclical inquiry.
- Assessment reform: Implement assessment reform by utilising I-TPACK to enhance current assessment methods. For example, cross-disciplinary teacher teams redesign assessments (e.g. hybrid AI-human essay grading) while documenting ethical trade-offs.

Future directions

Our proposed I-TPACK framework is new to policymakers, researchers, educators, and teacher educators. In this editorial, we suggest three research directions

- TPACK assessment metrics: further describe how each domain interacts with each other; develop validated rubrics to measure teacher progression across I-TPACK domains.
- Cross-cultural I-TPACK Adaptation: investigate how cultural, infrastructural, and policy contexts reshape domain prioritisation. Use comparative case studies analysing equity barriers and collaborative knowledge (HAIC-K) reconfigurations in diverse educational ecosystems.
- IPACK trajectories: Map evidence-based pathways for teacher development from AI literacy to competency. Design-based research testing micro-credential sequences (e.g. prompt engineering → ethics auditing → HAIC-K choreography) to optimise domain integration.

This editorial responds to urgent education issues – teacher education – and encourages us to have a conversation about the knowledge and skills needed for teacher education in the AI era. We look forward to learning more from your studies.

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